

The model's mental map is not a single model's quirk

Urban imageability and geographic knowledge in VLMs. Replication check, July 2026. 124 identical street-level points, no tools and no web search. Open-weight models run locally on Colab, so retrieval is not possible.

The July pilot found that a vision-language model, asked to locate a street scene, makes errors that are not random. Predictions are pulled toward the city centre (its cognitive "anchor"), and the pull is strongest where the urban fabric is most anonymous. This is the anchor-point distortion already described in the human cognitive-mapping literature. The obvious objection is that it might be one model's idiosyncrasy, so I ran the same 124 points through four independent model families.

1. Capability scales steeply with model tier

Model	family / origin	size	median error	≤25 km	country	city
gemini-3.5-flash	Google (proprietary)	—	9.7 km	77%	97%	72%
idefics2-8b	HuggingFace (Mistral-7B + SigLIP)	8B	39 km	44%	84%	52%
qwen2.5-vl-7b	Alibaba (open)	7B	495 km	36%	60%	24%
smolvlm-instruct	HuggingFace (open)	2B	—	0%	10%	3%

Geolocation ability drops sharply with model size. The 2B model essentially cannot place a street scene (country right only 10% of the time), so it is a capability floor rather than a usable comparison point. The mental-map test below therefore uses the three models that can locate.

2. The centripetal pull replicates across every capable model

Model	within-city responses	pulled toward centre	sign-test p	median pull toward centre
gemini-3.5-flash	216	58% (126/216)	0.017	0.47 km
qwen2.5-vl-7b	51	69% (35/51)	0.011	5.12 km
idefics2-8b	73	74% (54/73)	0.00005	6.28 km

Three independent model families, built by different teams (Google, Alibaba, and a Mistral-plus-SigLIP model from HuggingFace) on different data, all bend their errors toward the city centre, and all three effects are significant. The direction also fits the theory: **the frontier model pulls the least, and the two open models pull an order of magnitude harder** (about 5 to 6 km against Gemini's 0.5 km). That is what the anchor account predicts, since a model that extracts less visual evidence has to lean more on its prior for where the city sits. The bias is therefore not specific to one model. It looks like a by-product of learning to locate places from web imagery.

3. All three are pulled hardest where the fabric is most placeless

Urban fabric stratum	Gemini	Qwen	Idefics2
Commercial / business	50%	75%	56%
Historic-iconic core	56%	69%	69%
Dense residential	55%	44%	75%
Peripheral edge	60%	78%	77%
Low-density suburban	71%	75%	94%

In every model the anonymous periphery and suburb are pulled toward the centre most strongly (the suburban share is 71%, 75% and 94%), while the more distinctive commercial and historic cores hold their position better. The distortion follows exactly the urban-fabric strata the study samples, which is the imageability signal turning up inside three different artificial systems.

Scope and limits. This is pilot scale: 124 points, and the within-city samples for the open models are 51 (Qwen) and 73 (Idefics2), so their per-stratum figures rest on 10 to 16 points per cell. The effects are significant but the stratum gradient is a signal, not a settled result. SmolVLM-2B was too weak to test. Gemini ran on four separate views and the open models on a 2x2 grid, but the pull is measured within each model's own errors, so the presentation cannot create the effect on its own. The scaled study, 50 cities, about 10,000 points and 6 or more models, is what would turn this into a firm finding.

Every number is reproducible from the local database (PostgreSQL/PostGIS). The open-weight models run on Colab's free T4 with no tools and no network access to the models, so retrieval is not possible and the study measures the knowledge internal to each model.